

Chapter 1

Welcome to BenMAP-CE

In this chapter, find...

- **An overview of the tool.**
- **Instructions for installing the tool.**
- **Contacts, sources for more information, and answers to frequently asked general questions.**

The environmental Benefits Mapping and Analysis Program—Community Edition (BenMAP-CE) is a powerful, yet easy-to-use program that estimates the number and economic value of health impacts resulting from changes in air pollution concentrations. The open-source BenMAP-CE tool replaces the proprietary version of the program (BenMAP) that the U.S. Environmental Protection Agency (U.S. EPA) first developed in 2003 to analyze national-scale air quality policies. These analyses include health benefits assessments for the National Ambient Air Quality Standards (NAAQs) for Particulate Matter (2006, 2012) and Ozone (2008, 2010) as well as the Locomotive Marine Engine Rule (2008).

U.S. EPA and its partners designed BenMAP-CE to serve the analytical needs of a range of users, including scientists, policy analysts, and decision makers. Most users apply the BenMAP-CE tool to answer one of two types of questions:

1. What are the human health and economic benefits associated with a policy improving air quality?
2. What is the human health burden attributable to total air pollution levels?

While the BenMAP-CE development team designed the program to be accessible to novice users, the tool includes a number of features that will appeal to advanced analysts as well. For example, analysts can add their own health impact and valuation functions, map results, and perform a suite of sensitivity analyses. Beginning users can take advantage of U.S. EPA's pre-programmed settings and reports in the core program.

1.1 Overview of BenMAP-CE & Benefits Assessment

The BenMAP-CE program estimates the human health impacts and economic value of air quality changes. That is — BenMAP-CE relates air quality changes to human health benefits. Such analyses are a critical component of air quality policy assessments. As such, a variety of Federal, State and Local air pollution officials have used BenMAP-CE to inform air quality management decisions.¹

BenMAP-CE estimates benefits from improvements in human health, such as reductions in the risk of premature death, heart attacks, and other adverse health effects. Other benefits of reducing air pollution (i.e., visibility and ecosystem effects) are not quantified in the current version of BenMAP-CE. After estimating the reductions in the incidence of adverse health effects, BenMAP-CE calculates the monetary benefits associated with those reductions.

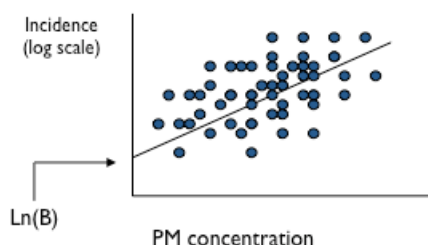
How does BenMAP-CE estimate human health effects?

First, BenMAP-CE determines the change in ambient air pollution using user-specified air quality data. Because BenMAP-CE does not model air quality changes, these data

¹ For a list of peer-reviewed articles that used the BenMAP and BenMAP-CE tools, see: www.epa.gov/benmap

must be input into BenMAP-CE as modeling data or generated from air pollution monitoring data (although some monitoring data is pre-loaded in BenMAP-CE, see Chapter 5 for details). Next, BenMAP-CE relates the change in pollution concentration with certain health effects (also known as health endpoints, see Chapter 6 for details). This relationship is often referred to as the health impact function (HIF) or the concentration-response (C-R) function. As shown in Figure 1-1, these HIFs are derived from epidemiology studies that relate pollutant concentrations with health outcomes. BenMAP-CE applies that relationship to the population experiencing the change in pollution exposure to calculate health impacts.

Epidemiology study



$$\text{Ln}(y) = \text{Ln}(B) + \beta(\text{PM})$$

Health impact function

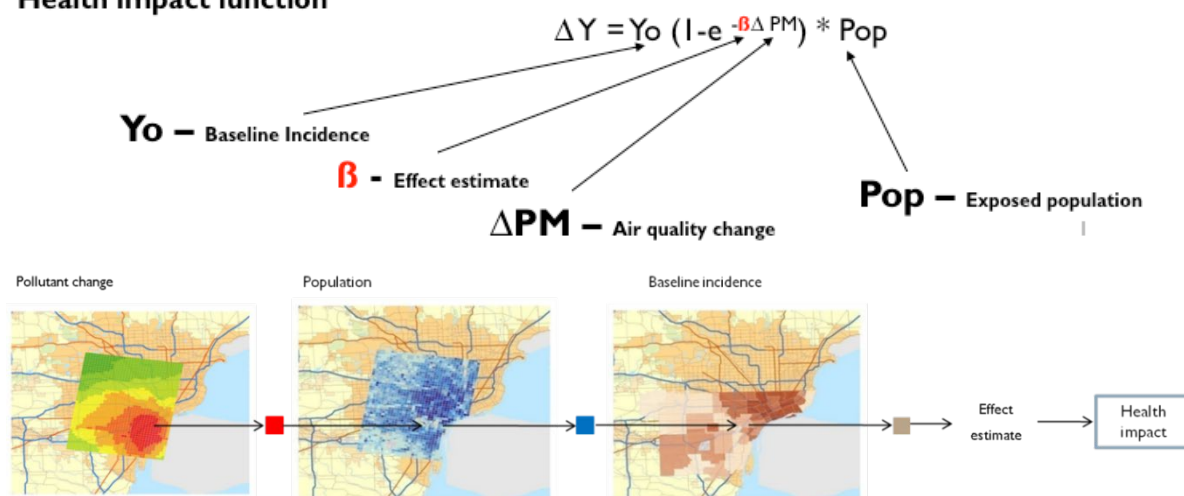


Figure 1-1. Deriving a Health Impact Function from the Epidemiology Literature

A simplified example is shown below.

Health Effect = Air Quality Change * Health Effect Estimate * Exposed Population *
Health Baseline Incidence

- **Air Quality Change.** The air quality change is the difference between the starting air pollution level (i.e., the baseline) and the air pollution level after some change, such as a new regulation (i.e., the control).
- **Health Effect Estimate.** The health effect estimate is an estimate of the percentage change in the risk of an adverse health effect due to a one unit change in ambient air pollution. Epidemiological studies are a good source for effect estimates.
- **Exposed Population.** The exposed population is the number of people affected by the air pollution reduction. The government Census office is a good source for this information. In addition, private companies may collect this information and offer it for sale.
- **Health Baseline Incidence.** The health incidence rate is an estimate of the average number of people who die (or suffer from some adverse health effect) in a given population over a given period of time. For example, the health incidence rate might be the probability that a person will die in a given year. Health incidence rates and other health data are typically collected by the government. In addition the World Health Organization is a good source for this.²

How does BenMAP-CE estimate the economic value of human health effects?

BenMAP-CE also calculates the economic value of avoided health effects (see Chapter 7 for details). After calculating the health changes, you can estimate the economic value by multiplying the reduction of the health effect by an estimate of the economic value per case (see Figure 1-2):

Economic Value = Health Effect * Value of Health Effect

² The World Health Organization is a good source for international health data, see: <http://www.who.int>.